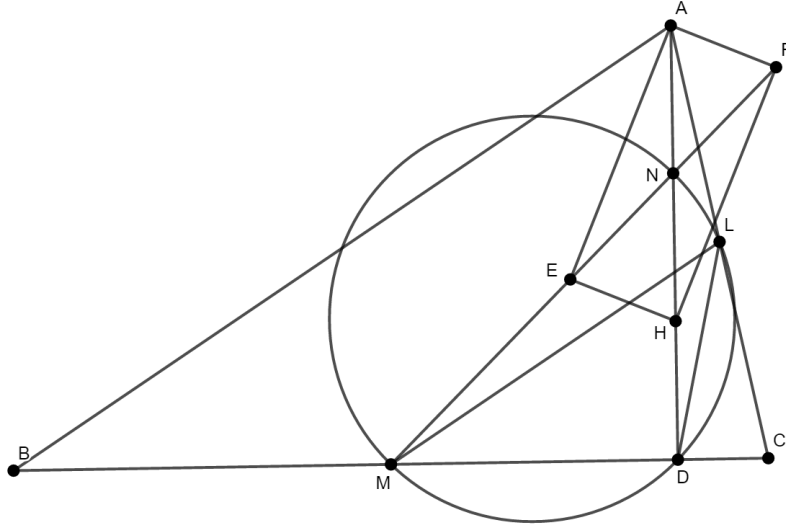


4.3.18



Let N, L, D be the midpoint of AH , the midpoint of AC , and the foot of A to BC . Since $AEHF$ is a rectangle, E, N, F are collinear. By nine-point circle, D, L, N, M are concyclic. Then

$$\begin{aligned}
 \angle ENH &= 2\angle EAH \\
 &= 2(\angle CAE - \angle CAD) \\
 &= \angle BAC - \angle CLD \\
 &= \angle CLM - \angle CLD \\
 &= \angle DLM \\
 &= \angle DNM
 \end{aligned}$$

so M, E, N are collinear, which implies F, E, M are collinear.